



**University Institute of Engineering**  
**Department of Computer Science & Engineering**

## **Experiment-5**

**Student Name:** Pranav Chauhan  
**Branch:** BE CSE (cyber security )  
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**UID:** 24BCY70004  
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1. **Aim of the practical:** To study and implement advanced SQL queries using **aggregate functions, conditional logic, joins, subqueries, and date operations** to solve real-world database problems involving **confirmation rates and trip cancellation rates**.

### 2. Questions

(A) A website maintains records of users who have registered on the platform and the status of confirmation requests sent to them. A confirmation request can either be successfully completed or may expire without completion.

For each registered user, calculate the **confirmation rate**, defined as:

**Confirmation Rate = Number of Successful Confirmations / Total Confirmation Requests**

if a user has never received any confirmation request, their confirmation rate should be considered **0**.

Display the **user ID** along with the **confirmation rate rounded to 2 decimal places**.

The result can be returned in any order.

| user_id | time_stamp             |
|---------|------------------------|
| 3       | 2020-03-21<br>10:16:13 |
| 7       | 2020-01-04<br>13:57:59 |
| 2       | 2020-07-29<br>23:09:44 |
| 6       | 2020-12-09<br>10:39:37 |

**Signups**

| user_id | time_stamp             | action    |
|---------|------------------------|-----------|
| 3       | 2021-01-06<br>03:30:46 | timeout   |
| 3       | 2021-07-14<br>14:00:00 | timeout   |
| 7       | 2021-06-12<br>11:57:29 | confirmed |
| 7       | 2021-06-13<br>12:58:28 | confirmed |
| 7       | 2021-06-14<br>13:59:27 | confirmed |
| 2       | 2021-01-22<br>00:00:00 | confirmed |
| 2       | 2021-02-28<br>23:59:59 | timeout   |

**Confirmations**



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**Solution:**

| user_id<br>[PK] integer | confirmation_rate<br>numeric |
|-------------------------|------------------------------|
| 2                       | 0.50                         |
| 3                       | 0.00                         |
| 6                       | 0.00                         |
| 7                       | 1.00                         |

```
select
    s.user_id,
    round(
        avg(case
            when action='confirmed' then 1
            else 0
        end),2) as confirmation_rate
from Signups s
left join Confirmations c
on s.user_id = c.user_id
group by s.user_id
order by s.user_id;
```

**(B)** A ride-booking company maintains information about trip requests and the users involved in those trips. Some users may be restricted from using the platform, while others are active participants. Each trip can either be completed successfully or cancelled by one of the participants.

For each day between **2013-10-01** and **2013-10-03**, determine the **proportion of canceled trips among all valid trips**. A trip should be considered valid only when both the customer and the driver associated with that trip are active users. The final result should display the **date and the calculated cancellation rate rounded to two decimal places**.

**Cancellation Rate = Cancelled Trips / Total Valid Trips**

Only consider trips where:

- Client is **NOT banned**
- Driver is **NOT banned**
- Date is between **2013-10-01** and **2013-10-03**



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### Users

| users_id | banned | role   |
|----------|--------|--------|
| 1        | No     | client |
| 2        | Yes    | client |
| 3        | No     | client |
| 4        | No     | client |
| 10       | No     | driver |
| 11       | No     | driver |
| 12       | No     | driver |
| 13       | No     | driver |

### Trips

| id | client_id | driver_id | city_id | status              | request_at |
|----|-----------|-----------|---------|---------------------|------------|
| 1  | 1         | 10        | 1       | completed           | 2013-10-01 |
| 2  | 2         | 11        | 1       | cancelled_by_driver | 2013-10-01 |
| 3  | 3         | 12        | 6       | completed           | 2013-10-01 |
| 4  | 4         | 13        | 6       | cancelled_by_client | 2013-10-01 |
| 5  | 1         | 10        | 1       | completed           | 2013-10-02 |
| 6  | 2         | 11        | 6       | completed           | 2013-10-02 |
| 7  | 3         | 12        | 6       | completed           | 2013-10-02 |
| 8  | 2         | 12        | 12      | completed           | 2013-10-03 |
| 9  | 3         | 10        | 12      | completed           | 2013-10-03 |
| 10 | 4         | 13        | 12      | cancelled_by_driver | 2013-10-03 |

### Solution:

| day<br>date | cancellation_rate<br>numeric |
|-------------|------------------------------|
| 2013-10-01  | 0.33                         |
| 2013-10-02  | 0.00                         |
| 2013-10-03  | 0.50                         |

```
SELECT
  t.request_at AS Day,
  ROUND(
    AVG(
      CASE
        WHEN t.status IN ('cancelled_by_driver', 'cancelled_by_client')
        THEN 1
        ELSE 0
      END
    ), 2
  ) AS cancellation_rate
FROM Trips t
JOIN Users c
  ON t.client_id = c.users_id
JOIN Users d
  ON t.driver_id = d.users_id
WHERE c.banned = 'No'
  AND d.banned = 'No'
  AND t.request_at BETWEEN '2013-10-01' AND '2013-10-03'
GROUP BY t.request_at
ORDER BY t.request_at;
```

### 3. Learning Outcomes:

- Learned to create and manage database tables using PostgreSQL.
- Performed data retrieval using SQL queries and joins.
- Applied aggregate functions and conditional statements to calculate rates.
- Improved practical skills in solving real-world database problems using SQL.
- Developed understanding of filtering and grouping data for meaningful results.